

Decadal soil and stand response to fire, harvest, and salvage-logging disturbances in the western boreal mixedwood forest of Alberta, Canada¹

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Abstract: Empirical knowledge of long-term ecosystem response to single and compound disturbances is essential for predicting disturbance effects and identifying management practices to maintain productive capacity of managed and restored landscapes. We report on soil, foliar nutrition, and regeneration growth response to wildfire, clearcut harvesting, and postfire salvage logging, as well as undisturbed control stands within the first year following disturbance and 10–11 years after disturbance in trembling aspen – white spruce mixedwood forests near Lesser Slave Lake, north-central Alberta, Canada. The compound disturbance of salvage logging resulted in greater long-term impacts on forest floor properties than either wildfire or harvesting alone. Changes in forest floor properties such as carbon and nitrogen pools and cation exchange capacity under salvage logging have persisted for 10 years and exhibit a different recovery trajectory than fire or harvesting. Forest floor properties under harvesting, including depth, carbon content, pH, extractable ammonium, and extractable sulphur, were not different from the control condition 10 years after harvest. Effects on soil and foliar nutrition were not reflected in productivity (height and diameter) of regenerating vegetation. Our results show differences between short- and long-term responses to disturbance, among single natural and anthropogenic disturbances, and among single and compound disturbances.

Key words: compound disturbance, salvage logging, soil carbon, soil nutrients, Boreal Plains Ecozone, boreal mixedwood forest.

Résumé : La connaissance empirique de la réponse des écosystèmes à des perturbations simples ou combinées est essentielle pour prédire les effets et identifier les pratiques d'aménagement pour maintenir la capacité productive des paysages aménagés et restaurés. Nous rapportons les effets du feu, de la coupe totale et de la coupe de récupération après feu, ainsi que leur comparaison avec des sites non perturbés, sur les sols, la nutrition foliaire et la croissance de la régénération, un an et 10–11 ans après perturbation dans des forêts mixtes de peuplier faux-tremble et d'épinette blanche près de Lesser Slave Lake (centre-nord de l'Alberta, Canada). Par rapport au feu ou à la coupe, la coupe de récupération après feu, qu'on peut qualifier de perturbation combinée, a causé des impacts plus importants à long terme sur les horizons organiques de surface. Les changements dans les propriétés des horizons organiques, tels que les réservoirs de carbone et d'azote et la capacité d'échange cationique, ont persisté après 10 ans, et montrent une trajectoire dans le temps différente de celles observées après feu ou coupe. Dix ans après perturbation, les propriétés des horizons organiques incluant l'épaisseur, le contenu en carbone, le pH et les concentrations en ammonium et soufre extractibles n'étaient pas différentes dans les sites coupés par rapport aux sites non perturbés. Les effets sur la nutrition des sols et des arbres ne se sont pas traduits par une plus grande productivité (hauteur et diamètre) de la régénération à ce stade précoce de la révolution du peuplement. Nos résultats montrent des différences entre la réponse à court et à long terme des écosystèmes suite à une perturbation, entre les perturbations d'origine naturelle et anthropique, et entre les perturbations simples et combinées dans les forêts boréales mixtes de l'Ouest.

Mots-clés : perturbation combinée, coupe de récupération, carbone du sol, éléments nutritifs du sol, Écozone de la Plaine Boréale, forêt boréale mixte.

Introduction

The western boreal mixedwood forest is experiencing an increasing anthropogenic footprint and an evolving suite of land-use pressures. It remains an important region of forest management for the sustainable production of solid wood, composite wood, and pulp products and for the communities supported by these activ-

ities (e.g., Comeau et al. 2005; Natural Resources Canada 2013), as well as being the focus of other resource extraction activities. Fire maintains a prominent role as a natural disturbance agent driving boreal forest regeneration and succession (Amiro et al. 2009; Brandt et al. 2013), but other natural disturbance agents such as insect pests, including bark beetles and defoliators, are also becoming more important (e.g., Volney and Fleming 2000; Safranyik

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Table 1. Forest floor properties compared by time of measurement, treatment, and their interaction.

		FF depth (cm)	pH	C (%)	N (%)	NH ₄ -N (µg·g ⁻¹)	NO ₃ -N (µg·g ⁻¹)
Time	<i>p</i> value	<0.000	<0.000	<0.000	<0.000	0.140	<0.000
I	Mean (SE)	5.14 (0.53)	5.58 (0.1)	41.58 (0.56)	1.59 (0.02)	96.32 (11.29)	12.58 (2.74)
P	Mean (SE)	8.19 (0.59)	6.38 (0.09)	38.47 (0.99)	1.83 (0.05)	87.28 (8.35)	115.5 (18.71)
Treatment	<i>p</i> value	0.044	0.001	0.006	0.532	0.139	0.239
C	Mean (SE)	10.20 (0.87)	5.62 (0.09)	42.26 (0.81)	1.7 (0.05)	62.41 (11.67)	23.57 (10.1)
B	Mean (SE)	2.97 (0.38)	6.74 (0.09)	38.12 (0.8)	1.7 (0.07)	134.1 (17.43)	64.64 (21.33)
H	Mean (SE)	10.21 (1.01)	5.18 (0.11)	48.1 (0.41)	1.84 (0.06)	111.5 (14.05)	101.3 (24.24)
S	Mean (SE)	2.69 (0.26)	6.82 (0.11)	34.61 (0.88)	1.56 (0.07)	59.25 (7.9)	49.37 (14.11)
Time × treatment	<i>p</i> value	0.028	0.011	0.052	0.052	<0.000	0.000
I × C	Mean (SE)	9.99 (1.28)	5.33 (0.10)	43.55 (0.95)	1.62 (0.06)	23.23 (3.31)	2.69 (0.18)
P × C	Mean (SE)	10.56 (0.94)	6.10 (0.08)	40.12 (1.35)	1.83 (0.09)	127.70 (23.92)	58.36 (25.27)
I × B	Mean (SE)	1.57 (0.13)	6.67 (0.12)	38.64 (0.88)	1.52 (0.04)	207.30 (22.83)	15.02 (6.00)
P × B	Mean (SE)	6.69 (0.90)	6.80 (0.13)	36.74 (1.79)	1.88 (0.09)	69.15 (13.32)	108.70 (37.36)
I × H	Mean (SE)	9.32 (1.41)	4.97 (0.14)	49.12 (0.56)	1.72 (0.06)	122.80 (20.98)	19.91 (6.85)
P × H	Mean (SE)	11.64 (1.32)	5.53 (0.15)	46.46 (0.29)	2.04 (0.10)	92.62 (13.00)	228.00 (48.04)
I × S	Mean (SE)	1.97 (0.29)	6.36 (0.12)	37.04 (0.50)	1.54 (0.12)	58.55 (15.39)	17.84 (6.07)
P × S	Mean (SE)	3.88 (0.39)	7.08 (0.12)	30.56 (1.84)	1.56 (0.09)	59.65 (9.19)	66.89 (20.74)

Note: Mean values and standard errors (parentheses) in bold (*p* values ≤ 0.05) are for interpretation, according to ANOVA results. Means of significant treatment C, carbon; N, nitrogen; CEC, cation exchange capacity; Exch Ca, Exch K, and Exch Mg, exchangeable calcium, potassium, and magnesium, respectively; Ext P and Ext S, (2012). Treatment: C, undisturbed control; B, burned; H, clearcut harvested; S, salvage logged.

et al. 2010; Price et al. 2013). As this disturbance mosaic evolves in extent, intensity, and complexity, empirical knowledge of long-term ecosystem response is essential for accurately projecting disturbance impacts and identifying appropriate management practices for maintaining the productive capacity of managed and restored landscapes.

Over the past two decades, there has been increasing interest in developing forest management approaches based on an understanding of natural disturbance dynamics (Bergeron et al. 2002). Large areas of the Canadian boreal forest were historically maintained naturally by stand-replacing wildfires (Brandt et al. 2013). To mimic the effects of wildfire, forest management approaches and silvicultural techniques have been designed to try to replicate the periodicity and spatial patterns of fire to maintain a spectrum of forest composition and structure at different scales in the landscape. However, wildfire and forest practices such as clearcutting affect more than the spatial configuration of the landscape, having unique effects on soils and biogeochemical processes at the stand and site levels (Maynard et al. 2014). These changes to ecosystem processes may push ecosystem functioning of harvested and wildfire-affected sites along different trajectories, with specific effects varying depending on site conditions (Thiffault et al. 2007; Rothstein and Spaulding 2010; Lorente et al. 2013). Therefore, the ecological significance of stand-level differences in soil biogeochemical patterns, processes, and trajectories between clearcutting and wildfire are important considerations in developing and improving natural disturbance based models of forest management.

There is emerging and increasing economic pressure in Canada's boreal forests to salvage merchantable timber from burned or insect-killed forests (Schmiegelow et al. 2006) as it is considered a promising feedstock source for future expansion of the bioenergy sector (Dymond et al. 2010). Compound disturbances occur when multiple disturbances happen in rapid succession and may result in changes to ecosystem recovery processes (Buma and Wessman 2011). Activities such as salvage logging may act as a compound disturbance by altering the ecosystem's recovery mechanisms. The potential impacts of salvage logging on soil biogeochemical processes and stand productivity have not received focused attention, in part because long-term ecosystem recovery processes after natural disturbance are still poorly understood and potential cumulative effects of natural and human disturbance have not been well documented (Lindenmayer and Noss 2006).

Wildfires usually remove the nutrient-rich crown material, understory, and part or all of the forest floor (Certini 2005) and may also have an impact on soil organic carbon of mineral layers (Chen and Shrestha 2012). However, most of the large woody material remains on the site (Wei et al. 1997; Tinker and Knight 2000; Johnson et al. 2005), likely playing a role in soil fertility and nutrient availability in the subsequent stand rotation (Marañón-Jiménez and Castro 2013), as well as carbon (C) accumulation. In the boreal forest of Quebec, on sites subject to high-severity fires, salvage logging (removal of merchantable timber following wildfire) has led to depleted soil calcium (Ca), magnesium (Mg), and phosphorus (P), which may not return to prefire levels within the planned rotation time of the stand (Brais et al. 2000). Reduced densities of saproxylic beetles in postfire salvage logging are implicated in complex shifts in soil nitrogen (N) availability and the growth of postfire-colonizing plant species (Cobb et al. 2010).

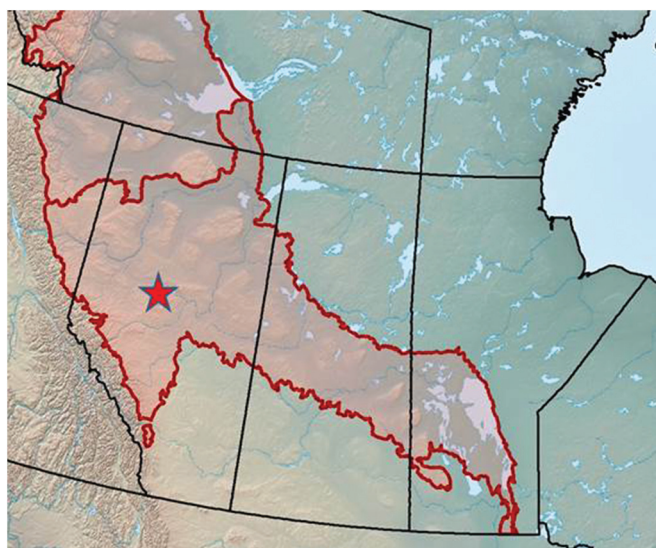
Much of our knowledge of impacts of disturbance on ecosystem functioning is based on studies of single disturbances (White 1979; Turner 2010). However, with multiple competing land uses, interactions between multiple disturbances are becoming of greater concern due to their potential for nonlinear behavior and long-lasting impacts on ecosystem structure and function (Buma and Wessman 2011). Interactions between multiple disturbances may cause novel disturbance characteristics (Buma and Wessman 2011) and unexpected recovery pathways (D'Amato et al. 2011). Given the potential for disturbances to cause large changes in ecosystem functioning, we require quantitative evaluation of the potential compounding effects of salvage logging in relation to the effects of single disturbances.

The boreal mixedwood forest of northern Alberta (Canada) is an ecosystem subject to multiple disturbances, over both time and space. For example, a large, severe wildfire (Chisholm Fire) occurred in May and June 2001, south of Lesser Slave Lake, in north-central Alberta. The fire burned 116 000 ha and resulted in the loss of 4.5 million m³ of growing stock (Chisholm Fire Review Committee 2001). The fire was exceptional in its extent, severity, and proximity to the hamlet of Chisholm. The fire occurred within a mosaic of white spruce (*Picea glauca* (Moench) Voss), trembling aspen (*Populus tremuloides* Michx.), black spruce (*Picea mariana* (Mill.) BSP.), and jackpine (*Pinus banksiana* Lamb.) within a region of

CEC	Exch Ca (cmol(+)-kg ⁻¹)	Exch K (cmol(+)-kg ⁻¹)	Exch Mg (cmol(+)-kg ⁻¹)	Ext P (μg·g ⁻¹)	Ext S (μg·g ⁻¹)	C content (kg·ha ⁻¹)	N content (kg·ha ⁻¹)
0.008	0.148	<0.000	0.223	<0.000	0.323	<0.000	<0.000
91.31 (2.34)	73.31 (3.83)	3.97 (0.19)	10.19 (0.59)	112.1 (6.11)	67.17 (3.35)	18928 (1829)	826.6 (71.74)
84.63 (2.82)	83.38 (3.38)	1.79 (0.08)	11.17 (0.54)	45.01 (3.25)	68.9 (3.24)	28677 (2199)	1401 (109.2)
0.985	0.488	0.000	0.543	0.108	0.748	0.073	0.104
90.24 (3.63)	77.63 (4.67)	2.32 (0.14)	12.07 (0.68)	83.82 (8.98)	63.2 (3.5)	38519 (1409)	1481 (138.9)
86.48 (3.19)	70.89 (5.86)	4.2 (0.41)	8.37 (0.8)	46.77 (4.41)	73.19 (5.79)	10604 (1266)	544.4 (65.11)
90.47 (3.22)	73.68 (3.35)	2.87 (0.17)	10.11 (0.5)	103.5 (9.71)	72.36 (3.89)	34020 (1999)	1599 (141.5)
83.25 (4.66)	96.65 (7.08)	2.04 (0.19)	12.64 (1.24)	81.95 (5.66)	62.41 (6.51)	10645 (1306)	623.6 (69.39)
0.01	<0.000	<0.000	<0.000	<0.000	<0.000	0.009	0.018
91.86 (4.54)	79.55 (7.67)	2.79 (0.17)	12.25 (0.92)	116.40 (10.27)	58.14 (4.32)	37786 (4001)	1342 (188.7)
87.54 (6.17)	75.61 (5.37)	1.82 (0.17)	11.88 (1.04)	29.50 (4.09)	71.64 (5.55)	38076 (4503)	1713 (191.1)
88.43 (4.09)	42.74 (2.77)	6.60 (0.17)	5.05 (0.45)	53.96 (8.04)	80.54 (10.91)	6445 (4403)	340.4 (61.9)
84.75 (4.87)	95.91 (6.46)	2.08 (0.15)	11.32 (1.04)	43.62 (4.93)	66.67 (4.93)	21695 (3330)	1089 (165.1)
89.33 (3.85)	72.38 (4.24)	3.65 (0.11)	9.92 (0.66)	148.10 (7.90)	60.53 (3.25)	29476 (2078)	1283 (161.2)
92.32 (5.80)	75.77 (5.57)	1.63 (0.12)	10.42 (0.78)	31.55 (2.64)	91.43 (6.64)	41340 (3415)	2108 (218.7)
100 (6.65)	117.50 (10.02)	2.85 (0.35)	15.82 (2.13)	87.83 (10.68)	92.19 (12.44)	8844 (1782)	558.5 (75.82)
73.93 (5.12)	86.24 (8.51)	1.64 (0.15)	11.05 (1.41)	78.69 (6.62)	45.86 (3.77)	13597 (2619)	696 (132.9)

effects were separated using Tukey's tests. Only variables for which there were significant time × treatment mean differences are shown in Fig. 2. FF, forest floor; extractable phosphorus and sulphur, respectively. Time: I, initial measurements after disturbance (2001 or 2002); P, measurements 10 or 11 years after disturbance

Fig. 1. Outline indicates the complete Boreal Plains Ecozone and a portion of the adjacent Taiga Plains Ecozone. The star indicates the study site location within the Boreal Plains Ecozone. (This figure is available in colour online.)



active forestry operations. Postfire salvage-logging activities were undertaken within a year following the fire, resulting in a mosaic of disturbances within the region.

This study was initiated to determine the short-term and longer term effects of fire, harvesting, and the compound disturbance of salvage logging on soil biogeochemical cycles and site productivity in the western boreal mixedwood forest of Alberta. The specific objectives were to (i) quantify and compare changes in carbon and nutrients of the forest floor and upper soil mineral layers and in foliar nutrition following severe wildfire, conventional clearcut harvesting, and postfire salvage logging relative to undisturbed controls in mixed trembling aspen – white spruce stands, (ii) compare initial (within 1 year) after disturbance with 10- or 11-year postdisturbance changes in those properties to establish recovery trajectories, and (iii) relate trajectories in soil fertility and foliar nutrition under these disturbance types to development and productivity of regenerating stands.

Material and methods

Study sites

Research plots were established in 2001 and 2002 near Chisholm, Slave Lake, and Canyon Creek, Alberta (Fig. 1). The research plots are in the Mid-Boreal Uplands EcoRegion within the Boreal Plains Ecozone (Wiken 1986; EcoRegions Working Group 1989), corresponding to the Lower Foothills Natural Subregion (Alberta Environmental Protection 1994). The research plots are located between 54°54.5'N and 55°42.0'N and between 114°4.42'W and 115°50.5'W. Soils are formed on glacial till and glacio-lacustrine deposits derived from sedimentary materials (La Roi 1992). Terrain is generally flat to rolling, and soils are Eutric, Melanic, and Dystric Brunisols and Dark Gray Luvisols with predominantly silt, silty loam, and silty clay loam surface textures (Soil Classification Working Group 1998), with surface soil texture relatively consistent across the study area.

Prior to disturbance, stand cover types were white spruce – deciduous mixedwoods and contained between 60% and 40% white spruce and between 40% and 60% trembling aspen or balsam poplar. Only sites deemed productive based on composition (i.e., 51%–70% crown closure), density (“C” grade density; Alberta Forestry, Lands, and Wildlife 1991), and size of trees (i.e., >25 m) in the stand were selected for the study.

Plots were located on upland boreal mixedwood sites that are mesic moisture regime and medium nutrient regime (Beckingham and Archibald 1996), with low-bush cranberry, feather moss, and wild rose present on all sites in varying abundance. Consistency in overstory cover type, surface soil texture, ecosite, and drainage were the basis of site selection. Due to the nature of the wildfire and harvesting operations, replicates within a treatment were, in some cases, tens of kilometres apart but consistent with respect to vegetation, site, and soil properties. The minimum distance between replicates was 400 m.

Plots were established in four disturbance types (treatments): (1) undisturbed control, (2) burned, (3) clearcut harvesting, and (4) postfire salvage logging. Each of the four disturbance types was replicated three times for a total of 12 research plots of approximately 2 ha each. Clearcut sites that subsequently burned exist within the region, however, not on a scale and pattern that would have allowed for replicated incorporation into the trial. Burned sites were within the area affected by the Chisholm fire of May

Fig. 2. Effects of time (initial measurements after disturbance (initial) and measurements at 10–11 years after disturbance (10/11 yr)) and treatment (undisturbed control, burned, clearcut harvested, and salvage logged) in forest floor properties in western boreal mixedwood forest stands of Alberta: (A) forest floor (FF) depth; (B) pH; (C) extractable $\text{NH}_4\text{-N}$; (D) extractable $\text{NO}_3\text{-N}$; (E) extractable P; (F) extractable S; (G) cation exchange capacity (CEC); (H) exchangeable Ca; (I) exchangeable K; (J) exchangeable Mg; (K) C content; and (L) N content. Different lowercase letters indicate significant differences ($P \leq 0.05$; Tukey's tests) among time $\times$ treatment interactions.

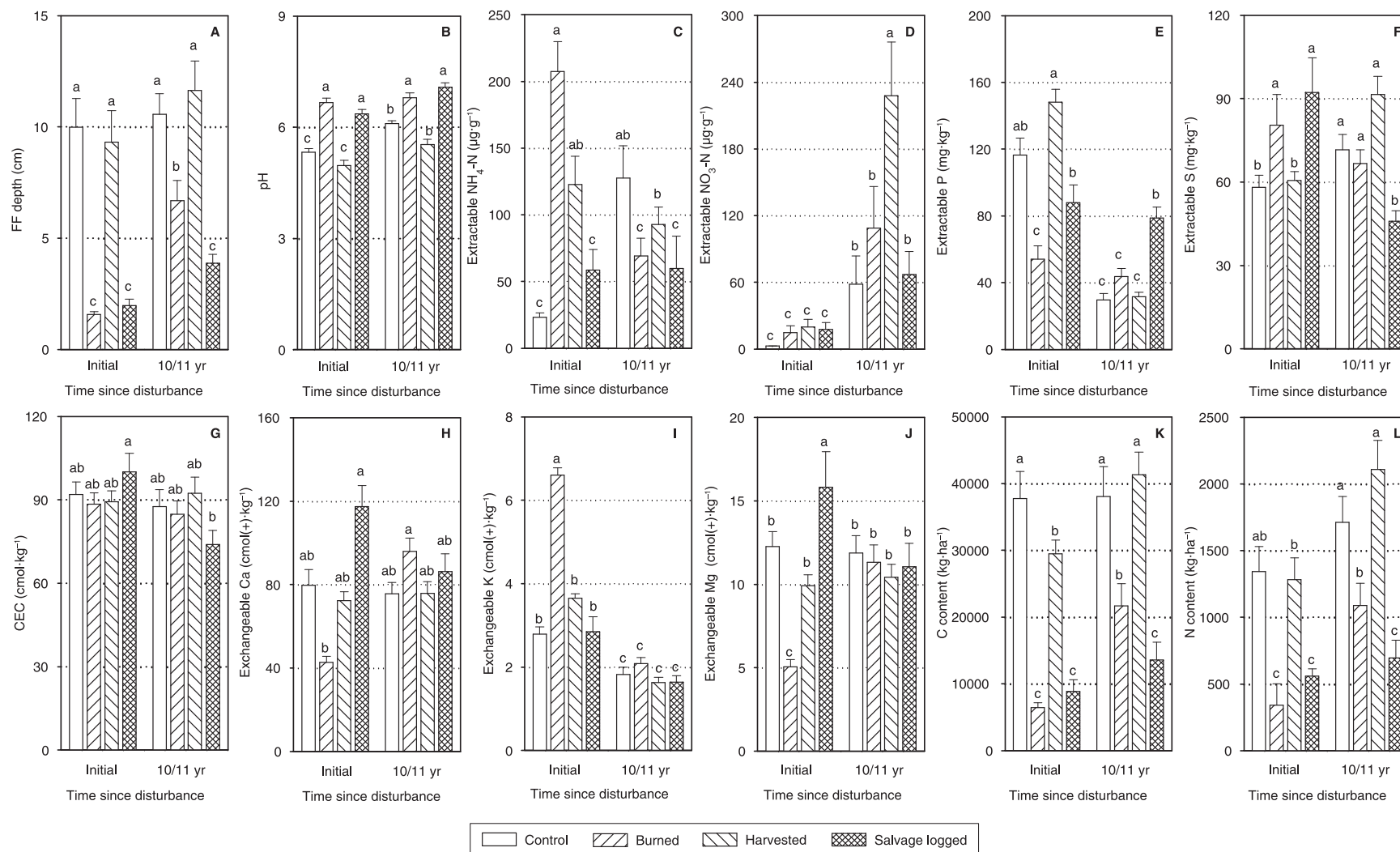
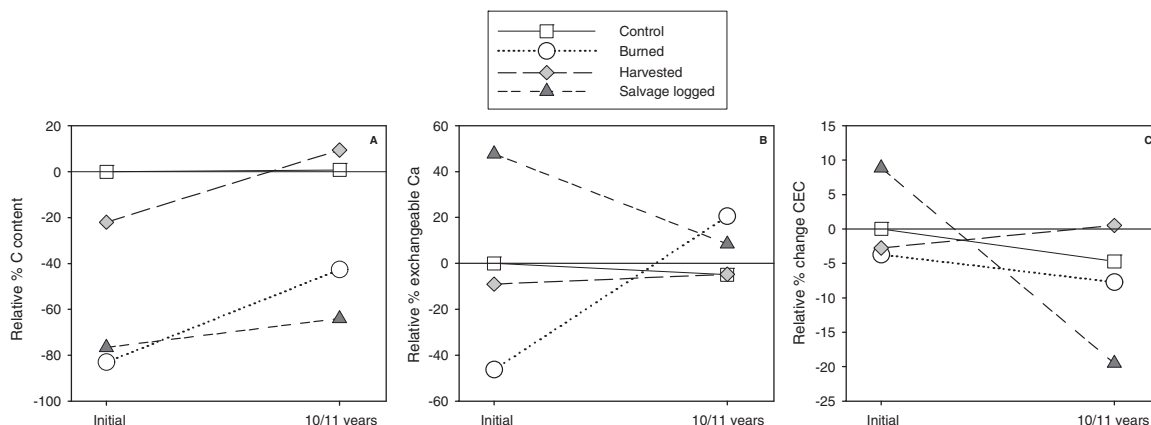


Fig. 3. Percentage change from initial conditions on the control plots on soil properties ((A) forest floor C content; (B) exchangeable Ca; (C) cation exchange capacity (CEC)) for initial postdisturbance measurements (initial) and measurements at 10 or 11 years after disturbance (10/11 years) following four disturbance types (undisturbed control, burned, clearcut harvested, and salvage logged).



2001 and had not previously been burned in other recent fires in the area. Plots were established in burned areas in July 2001. Plots in undisturbed areas were established in July and August 2002. Clearcut sites were harvested over the winter of 2001–2002, and plots were established in July and August 2002. Salvage-logged sites were harvested in the winter of 2001–2002 following the Chisholm fire, and plots were established in July and August 2002. Harvesting was done with a tracked feller–buncher, rubber-tired grapple skidder, and roadside processor. Care was taken so that plots with different treatments were interspersed in space, within the limits set by the configuration of the burned area. The 2 ha research plots contained a soil sampling zone of approximately 1 ha, as well as a surrounding area used for associated studies on coarse woody debris and invertebrate assemblages reported on elsewhere (Smith et al. 2008; Cobb et al. 2010).

Soil sampling and analysis

Soils were initially sampled within the first year following disturbance: in July 2001 for burned plots (approximately 6 weeks after fire) and in July 2002 for clearcut and salvage-logged plots (approximately 5 months after harvest) and control plots. This sampling time is hereafter referred to as “initial after disturbance” or simply “initial”. Plots were resampled again in late June 2012, i.e., 10 or 11 years after disturbance. Soils were sampled from a random selection of points within a systematic grid layout, with grid intersections located at 20 m × 20 m intervals in each direction and each grid containing 16 sampling points. Grids were positioned to maximize the effective sampling area for the shape of the plot and were established consistently by cardinal directions. An area of approximately 1 ha (50 m width) remained around each sampling grid as a buffer area. Soils were sampled at six randomly chosen grid points of the 16 within each plot, with the same grid points used in the initial after disturbance and 10 or 11 years postdisturbance sampling.

Separate forest floor (combined organic layer of litter (L), fibric (F), and humic (H) horizons) samples, excluding surface woody debris, were collected for chemical and mass analysis at each sampling location. Forest floor material was completely removed by the fire at some sampling points; where no forest floor was present at a given sampling location, it was recorded as a zero within the plot mean. Forest floor depth at four corners of a 15 × 15 cm quadrat was measured and averaged to provide a mean forest floor depth for each sampling location. Forest floor mass was determined by removing the complete forest floor within the quadrat. Oven-dried mass of the forest floor was used to determine bulk density, which was in turn used to calculate elemental content (kilograms per hectare). Surface mineral soil (0–7 cm

depth) samples were taken at each sampling location for chemical analysis, and a separate 500 cm³ soil core using a sliding-hammer corer was taken to determine bulk density. The 0–7 cm depth included Ah, Ae, and AB horizons. Samples for chemical analysis were kept in coolers with ice packs until transport to Edmonton and then refrigerated (3 °C) until analysis. Forest floor samples were homogenized in a blender while moist, with subsamples air-dried for analyses as required. Mineral soil samples were air-dried, sieved to 2 mm, and ground as required for analysis.

Samples were analyzed according to the methods of Kalra and Maynard (1991). Forest floor samples were analyzed for pH (1:4 ratio of soil to liquid in 0.01 mol·L⁻¹ calcium chloride (CaCl₂)); total C was determined by LECO analysis; total nitrogen (N) was determined by the Kjeldahl method; and extractable NH₄-N and NO₃-N were determined on fresh samples in 2 mol·L⁻¹ KCl. Extractable P was extracted using Bray P-1 extractant, sulphur (S) was determined using 0.01 mol·L⁻¹ NH₄Cl, and cation exchange capacity (CEC), exchangeable Ca, magnesium (Mg), potassium (K), and sodium (Na) were determined in unbuffered ammonium chloride with elemental analysis by inductively coupled plasma atomic absorption spectrometry. Mineral soils were analyzed for pH in a 1:2 ratio of soil to liquid (0.01 mol·L⁻¹ CaCl₂), total C and N and extractable P were determined as for the forest floor, S was determined using Ca(H₂PO₄)₂ (500 mg·L⁻¹ P), and CEC, exchangeable Ca, magnesium Mg, potassium K, and sodium Na were determined as for the forest floor. Particle size analysis was done on mineral soil samples to determine surface soil texture.

Foliar sampling and analysis

Foliage was sampled from clonally regenerating trembling aspen in plots in burned, clearcut harvested, and salvage-logged treatments in June 2003 and again in June 2012. Trees were 2 years old at the first date of sampling. In 2012, foliar samples were taken from trees adjacent to three (randomly selected) of the six soil sampling locations within a replicate, creating a composite sample of three trees per replicate. Leaves were taken from the upper one-half to one-third of the tree crown of individual trees as close as possible to the soil sampling locations. Sufficient leaves were taken to loosely fill a 500 mL container. Samples were kept cool in paper bags until oven dried at 70 °C for 24 h and then ground prior to analysis. Foliage was not sampled in control stands as all trembling aspen trees were mature overstory trees rather than regeneration trees.

Foliage samples were analyzed for total N by Kjeldahl N (Kalra and Maynard 1991). Samples were digested in nitric acid and analyzed for P, K, S, Ca, Mg, manganese (Mn), iron (Fe), copper (Cu),

Table 2. Mineral soil properties at a depth of 0–7 cm compared by time of measurement, treatment, and their interaction.

		pH	C (%)	N (%)	NH ₄ -N (µg·g ⁻¹)	NO ₃ -N (µg·g ⁻¹)	CEC
Time	<i>p</i> value	<0.000	0.273	0.482	0.8	0.138	0.715
I	Mean (SE)	5.46 (0.11)	3.12 (0.18)	0.23 (0.03)	11.72 (2.09)	7.33 (1.06)	21.16 (2.79)
P	Mean (SE)	6.01 (0.12)	2.84 (0.32)	0.27 (0.04)	7.04 (0.41)	10.23 (2.18)	23.52 (3.64)
Treatment	<i>p</i> value	0.018	0.373	0.573	0.146	0.531	0.646
C	Mean (SE)	5.04 (0.12)	3.84 (1.05)	0.43 (0.08)	4.64 (0.94)	13.58 (3.66)	40.99 (8.31)
B	Mean (SE)	6.39 (0.13)	2.18 (0.15)	0.16 (0.01)	19.72 (3.22)	8.33 (2.14)	13.05 (0.71)
H	Mean (SE)	4.56 (0.13)	2.84 (0.25)	0.23 (0.05)	4.41 (0.49)	6.69 (0.9)	18.99 (3.19)
S	Mean (SE)	6.29 (0.14)	3.54 (0.43)	0.2 (0.02)	4.31 (0.75)	5.8 (0.48)	18.02 (1.55)
Time × treatment	<i>p</i> value	0.315	0.844	0.999	<0.000	0.343	0.704
I × C	Mean (SE)	4.76 (0.15)	43.55 (0.95)	1.62 (0.06)	0.45 (0.07)	9.31 (0.84)	42.60 (11.19)
P × C	Mean (SE)	5.50 (0.17)	40.12 (1.35)	1.83 (0.09)	9.53 (1.28)	18.80 (8.01)	38.32 (12.35)
I × B	Mean (SE)	6.30 (0.17)	38.64 (0.88)	1.49 (0.06)	25.04 (4.19)	8.67 (2.96)	12.59 (0.84)
P × B	Mean (SE)	6.66 (0.18)	36.74 (1.79)	1.88 (0.09)	5.54 (0.24)	7.53 (1.73)	14.28 (1.33)
I × H	Mean (SE)	4.19 (0.15)	49.12 (0.56)	1.72 (0.06)	2.45 (0.49)	5.66 (0.20)	16.19 (3.10)
P × H	Mean (SE)	5.13 (0.18)	46.46 (0.29)	2.04 (0.10)	6.57 (0.54)	8.25 (2.22)	23.04 (6.40)
I × S	Mean (SE)	6 (0.17)	37.04 (0.50)	1.54 (0.12)	2.67 (1.13)	5.46 (0.27)	17.76 (1.96)
P × S	Mean (SE)	6.77 (0.19)	30.56 (1.84)	1.56 (0.09)	6.51 (0.58)	6.36 (1.21)	18.45 (2.61)

Note: Mean values and standard errors (parentheses) in bold (*p* values ≤ 0.05) are for interpretation, according to ANOVA results. Means of significant treatment N, nitrogen; CEC, cation exchange capacity; Exch Ca, Exch K, and Exch Mg, exchangeable calcium, potassium, and magnesium, respectively; Ext P and Ext S, extractable Treatment: C, undisturbed control; B, burned; H, clearcut harvested; S, salvage logged.

and zinc (Zn) using inductively coupled plasma atomic absorption spectrometry.

Tree mensuration

Overstory and regenerating tree mensuration was completed in 2012 on the same randomly selected subset of three plots for each replicate in control, burned, clearcut harvested, and salvage-logged treatments. Mensuration results for regeneration in the latter three treatments are reported here. A minimum sample size of 25 regenerating trembling aspen was required for each plot sampled. To facilitate the minimum sample size, plot radii of 1.78 m or 3.99 m were used based on regenerating trembling aspen density. Species and diameter at breast height (DBH) were recorded for all trees with a total height greater than 50 cm, and the total height of all white spruce and the three tallest trembling aspen were measured.

Statistical analysis

Soil, foliar, and mensuration data were analyzed as a two-factor experiment arranged in a completely randomized design using mixed linear models (Littell et al. 1996). The two fixed factors were time of measurement (initial after disturbance or 10 or 11 years after disturbance) and treatment (undisturbed control, burned, clearcut harvested, and salvage logged for soil properties and burned, clearcut harvested, and salvage logged for foliar and mensuration properties). We verified the absence of temporal autocorrelation between samples immediately after disturbance and 10 or 11 years after disturbance by adding a dependence parameter using a first-order regressive model (AR1) (Pinheiro and Bates 2000). Due to the spatial structure of the data (i.e., samples from the same plot are more likely to be similar), the plot was defined as a random factor. Significant differences were tested using analysis of variance (ANOVA), with post hoc Tukey's multiple comparison tests to determine significant differences among treatments.

Plots of fitted and observed values and residuals were examined for deviations from normality assumptions (Pinheiro and Bates 2000); data did not require transformation to meet the assumptions of the analyses. Analyses were conducted in SAS (version 9.2, SAS Institute Inc., Cary, North Carolina), using PROC MIXED and PROC ARIMA. Statistical significance was declared at $\alpha = 0.05$.

Results

Forest floor

All forest floor properties were significantly influenced either by time of measurement (initial after disturbance vs. 10 or 11 years after disturbance), treatment (control, burned, clearcut harvested, or salvage logged), and (or) their interaction (Table 1). Both time and treatment had a significant effect on forest floor total C concentration; initial values after disturbance were higher compared with those measured 10 or 11 years after disturbance, and values under harvesting were higher than those under burning and salvage logging (Table 1). Total N concentration was only significantly affected by time, with the highest values measured 10 or 11 years after disturbance (Table 1). All other variables (forest floor depth; pH; extractable NH₄-N and NO₃-N; CEC, exchangeable Ca, K, and Mg; extractable P and S; and C and N content) showed significant interactions between treatment and time.

Forest floor depth was consistently lower under salvage logging relative to other treatments at both points in time. Depth under burning alone was initially similar to salvage logging but increased significantly after a decade. Both control and harvested treatments had similar forest floor depth, which was significantly higher relative to other treatments at all points in time (Fig. 2).

Forest floor pH was similar in the undisturbed control and under harvesting and consistently lower than the fire-related treatments at both points in time but increased significantly between the measurement periods. Forest floor pH under burning and salvage logging was similar but did not change over time (Fig. 2).

Forest floor extractable NO₃-N was initially similar among treatments; it increased at 10 or 11 years, with concentration under harvesting increasing significantly more than for the other treatments (Fig. 2). Forest floor extractable NH₄-N concentration was initially lowest in the control and salvage-logged treatments and highest in the burned and harvested treatments. Extractable NH₄-N did not change through time under harvesting and salvage logging, whereas it significantly increased over time in the undisturbed stands and significantly decreased under burning (Fig. 2).

Forest floor extractable P was initially highest in the undisturbed control and under harvesting; it decreased after 10 or 11 years to become similar to the burned treatment, which did not change over time. Extractable P concentration did not change

Exch Ca (cmol(+)-kg ⁻¹)	Exch K (cmol(+)-kg ⁻¹)	Exch Mg (cmol(+)-kg ⁻¹)	Ext P (μg·g ⁻¹)	Ext S (μg·g ⁻¹)	C content (kg·ha ⁻¹)	N content (kg·ha ⁻¹)
0.011	<0.000	0.5	0.011	0.003	0.169	0.010
12.86 (1.04)	0.92 (0.05)	3.03 (0.38)	48.22 (3.59)	43.37 (2.08)	19429 (1063)	1057 (52.39)
19.43 (3.46)	0.41 (0.03)	2.91 (0.53)	35.56 (2.96)	35.15 (1.12)	18672 (1105)	1272 (66.65)
0.795	0.12	0.695	0.177	0.401	0.336	0.966
18.93 (4.76)	0.42 (0.03)	5.48 (1.15)	26.34 (2.18)	37.68 (1.95)	24552 (2191)	1287 (109.1)
13.18 (1.35)	1.16 (0.08)	2.07 (0.21)	53.21 (3.94)	39.12 (2.07)	14298 (764.4)	1033 (55.06)
11.84 (2.66)	0.6 (0.04)	2.23 (0.34)	28.09 (2.94)	36.04 (1.43)	19026 (1377)	1061 (80.65)
17.19 (1.85)	0.64 (0.06)	2.49 (0.3)	63.53 (7.98)	49.62 (4.81)	21298 (1872)	1180 (93.63)
0.002	<0.000	0.839	0.513	0.015	0.049	0.005
10.37 (1.31)	0.48 (0.04)	5.61 (1.49)	29.53 (3.12)	38.55 (2.64)	26492 (2833)	1367 (156.8)
33.19 (11.96)	0.30 (0.04)	5.27 (1.85)	21.02 (2.14)	36.22 (2.84)	21061 (3344)	1154 (127.4)
12.44 (1.66)	1.45 (0.07)	2.08 (0.27)	55.36 (5.17)	41.47 (2.76)	13124 (767.6)	914.7 (49.04)
15.16 (2.19)	0.38 (0.05)	2.04 (0.26)	47.49 (4.25)	32.84 (0.78)	17614 (1766)	1347 (129.6)
10.35 (2.61)	0.69 (0.05)	2.13 (0.30)	30.37 (4.15)	35.76 (0.94)	19641 (2037)	878.2 (79.91)
14.16 (5.53)	0.47 (0.05)	2.39 (0.76)	24.55 (3.82)	36.46 (3.40)	17951 (1330)	1345 (142.9)
18.37 (2.53)	0.72 (0.08)	2.81 (0.41)	72.14 (11.57)	58.34 (7.26)	23213 (2695)	1142 (125.8)
15.23 (2.56)	0.50 (0.09)	1.96 (0.39)	49.19 (8.36)	35.07 (0.59)	18320 (2214)	1242 (138.8)

effects were separated using Tukey's tests. Only variables for which there were significant time × treatment mean differences are shown in Fig. 3. C, carbon; phosphorus and sulphur, respectively. Time: I, initial measurements after disturbance (2001 or 2002); P, measurements 10 or 11 years after disturbance (2012).

through time under salvage logging, and at the 10 or 11 year measurement, it was highest among all treatments (Fig. 2).

Despite initial differences, forest floor extractable S converged to similar values at 10 or 11 years after disturbance in the control, burned, and harvested treatments. However, S concentration under salvage logging, which was initially highest among all treatments (along with burning), decreased to a significantly lower concentration after a decade relative to the other treatments (Fig. 2).

Cation exchange capacity was similar among all treatments and times of measurement, except under salvage logging, which displayed a significant decrease in CEC between the initial and 10 or 11 year measurements. At 10 or 11 years after disturbance, concentrations of exchangeable base cations (Ca, Mg, K) converged to similar levels among treatments despite initial differences between treatments. In the case of exchangeable K, concentration was lower at 10 or 11 years relative to initial measurements (Fig. 2).

Forest floor total C content was initially highest in the control and lowest in the salvage-logging treatment, with C content under salvage logging not changing significantly over time. In contrast, total C content under both burning and harvesting significantly increased over the decadal period, with forest floor C content under harvesting reaching similar levels to the control after 10 years. Forest floor total N content was also initially highest in the control treatment and did not change over time. For the other treatments, total N content significantly increased through time, with a convergence to the levels of the control treatment 10 or 11 years after disturbance (Fig. 2).

Long-term trajectories with initial control values set to zero are shown in Fig. 3, demonstrating percent change among treatments and over time for forest floor C content (kilograms per hectare) (Fig. 3A), forest floor exchangeable Ca (Fig. 3B), and forest floor CEC (Fig. 3C). Figure 3 plots the eight means representing the four treatments and two times of measurement, with no comparison of slopes for treatment, time, or interaction effects, as those comparisons are presented elsewhere (Table 1; Fig. 2). The temporal trends in Fig. 3 are shown as visual indicators of current recovery trajectories.

Mineral soil

Neither time of measurement nor treatment had a significant effect on CEC, total C, total N, extractable NO₃-N, and exchangeable Mg concentrations in the mineral soil (Table 2). However, values for pH and extractable P differed significantly with time;

highest values occurred immediately after disturbance for extractable P, whereas they occurred at 10 or 11 years after disturbance for pH. Treatments also differed significantly for pH; values were higher under both burned and salvage-logged plots compared with harvested plots, whereas they did not differ from the undisturbed control (Table 2).

All other variables (extractable NH₄-N, extractable S, exchangeable Ca and K concentrations, and total C and N contents) showed significant interactions between times of measurement and treatments (Table 2). For all these variables, although there were some differences between treatments at the initial measurement, all treatments converged to similar values after 10 or 11 years. Moreover, even at the initial measurement, differences between treatments were not large and most were not significant (Fig. 4).

Foliar nutrition and growth of regenerating stands

Foliar Ca, P, S, and Cu concentrations changed significantly through time (Table 3). Values were significantly higher immediately after disturbance relative to the 10 or 11 year measurements for foliar P, S and Cu, whereas the inverse was true for foliar Ca. Regardless of time, treatments differed significantly for foliar N (Table 3), with concentrations being higher under harvesting compared with both burning and salvage logging (Fig. 5). Trembling aspen regeneration DBH and height did not differ among treatments 10 or 11 years after disturbance (Fig. 5).

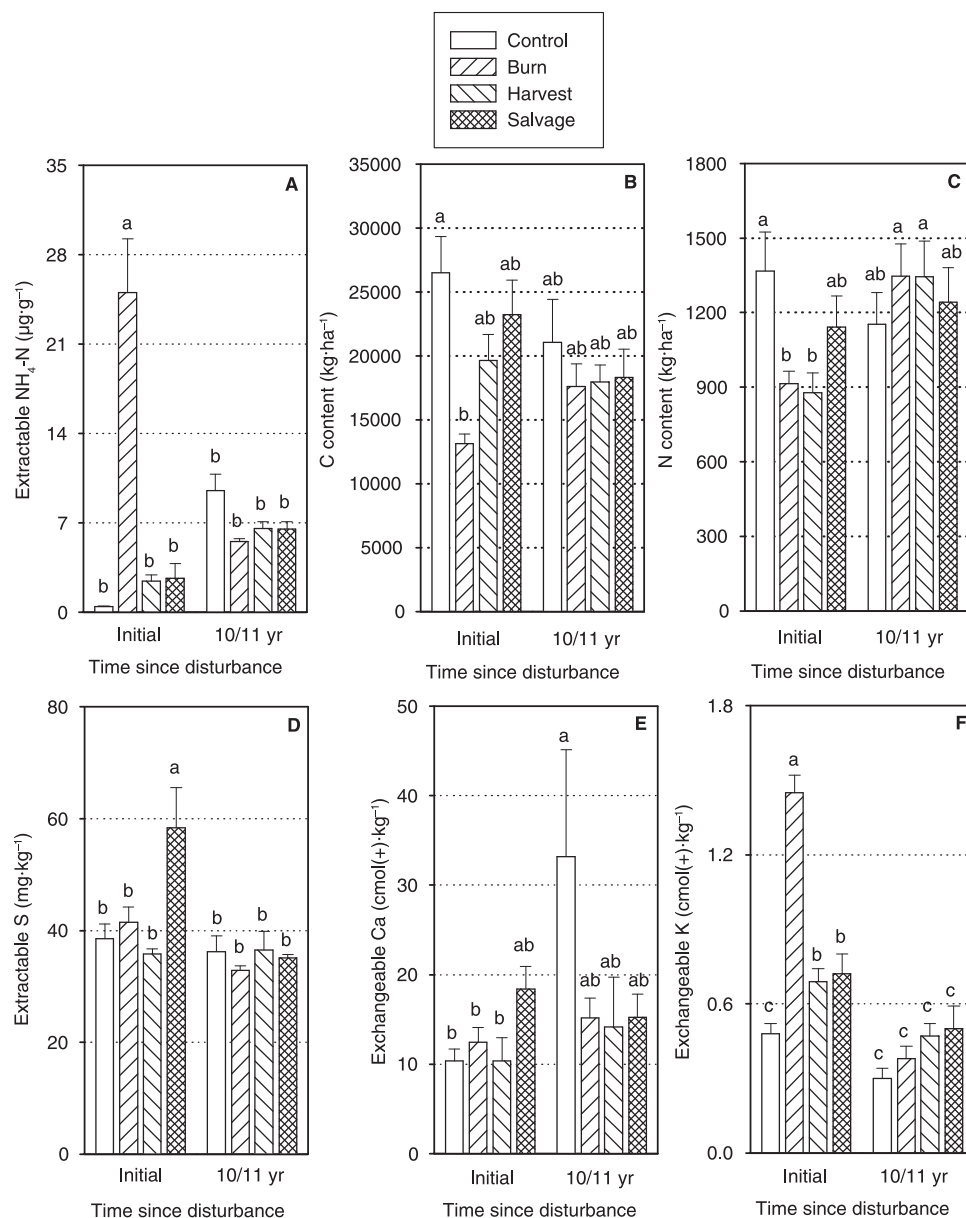
Discussion

Incremental effects of postfire salvage logging

This study has identified that the compound disturbance of post-fire salvage logging produced impacts on forest floor biogeochemistry that were cumulatively greater than the impacts of either individual fire or clearcut-harvest disturbances. Salvage logging also resulted in a recovery trajectory that was different from either the fire or harvesting disturbances. Disturbance type effects and recovery trajectories were most pronounced in the forest floor horizon.

A fundamental question addressed in this study was the comparative degree of forest floor change under fire and harvesting and the compounding effect of postfire salvage logging. The forest floor was almost completely removed by fire over large portions of this area, which, by itself, could potentially impact long-term site productivity through loss of soil organic matter and altering

Fig. 4. Effects of time (initial measurements after disturbance (initial) and measurements at 10–11 years after disturbance (10/11 yr) and treatment (undisturbed control, burned, clearcut harvested, and salvage logged) on properties of mineral soil (0–7 cm depth) ((A) extractable $\text{NH}_4\text{-N}$; (B) C content; (C) N content; (D) extractable S; (E) exchangeable Ca; (F) exchangeable K) in western boreal mixedwood forest stands of Alberta. Different lowercase letters indicate significant differences ($P \leq 0.05$; Tukey's tests) among time $\times$ treatment interactions.



short- and long-term nutrient supply (Certini 2005). On the other hand, postfire stands usually contain high stocks of deadwood C, contributing to total ecosystem C stocks (Seedre et al. 2011). Wild-fire has also been shown to increase soil C and N over time due to the sequestration of charcoal and organic matter forms and the natural invasion of N-fixing vegetation (Johnson and Curtis 2001). Woody debris present after disturbance is thought to become incorporated into soils of cool coniferous forests, contributing to soil C pools (Johnson and Curtis 2001). The forest floor C pool increased in all treatments over the 10-year period as expected under vegetation establishment and detrital inputs from woody material, litter, and roots (Finér et al. 1997; Laiho and Prescott 1999; Seedre et al. 2011). However, decadal recovery of C pools was slower under salvage logging than under either burning alone or harvesting (Fig. 3A). The N pool initially reduced under fire has recovered under both burn alone and salvage logging but re-

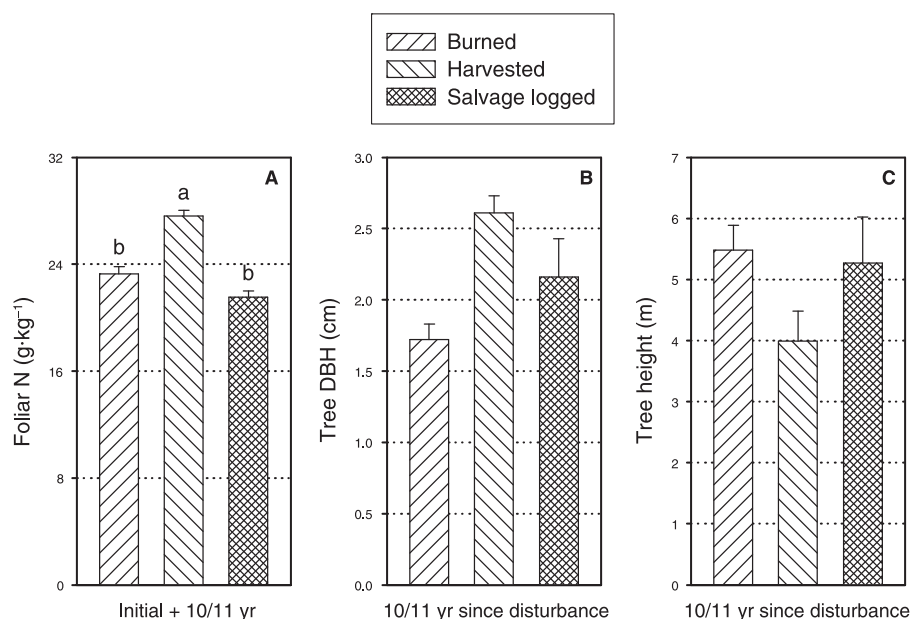
mained lower in the salvage-logging treatment than harvesting. Forest floor C and N on fire-affected sites and harvested sites have also been shown to converge to similar levels over time in boreal mixedwood stands (Shrestha and Chen 2010). Slower forest floor and elemental recovery under salvage logging than burn alone is presumably due to the initial additional removal of woody material in salvage logging that would have contributed to forest floor accumulation and development. However, it may also be due to differential rates of organic matter loss through decomposition and accumulation by aboveground and belowground detrital inputs over the decade (Shrestha and Chen 2010).

Nitrogen dynamics were inconsistently affected by disturbance, and the recovery trajectory differed among the disturbances. For example, extractable $\text{NO}_3\text{-N}$ levels were initially low and not altered by disturbance, consistent with other western boreal mixedwood sites (Kishchuk et al. 2014). Contrary to expectation (e.g.,

Table 3. Foliar nutrient concentrations ($\text{g}\cdot\text{kg}^{-1}$) in regenerating trembling aspen (*Populus tremuloides* Michx.) compared by time of measurement, treatment, and their interaction.

		Nutrient concentration ($\text{g}\cdot\text{kg}^{-1}$)							
		N	Mg	K	Ca	P	S	Mn	Cu
Time	<i>p</i> value	0.751	0.298	0.707	0.000	<0.000	<0.000	0.579	<0.000
	Mean (SE)	24.08 (0.44)	2.12 (0.06)	9.72 (0.17)	8.26 (0.23)	2.39 (0.05)	1.87 (0.04)	0.08 (0)	0.07 (0)
	Mean (SE)	24.31 (0.61)	2.3 (0.11)	9.95 (0.49)	10.45 (0.49)	1.79 (0.05)	0.2 (0.01)	0.08 (0.01)	0.01 (0)
Treatment	<i>p</i> value	0.002	0.708	0.694	0.094	0.185	0.396	0.586	0.369
	Mean (SE)	23.26 (0.57)	2.08 (0.06)	10.36 (0.26)	9.05 (0.33)	2.15 (0.08)	1.38 (0.11)	0.07 (0.01)	0.07 (0.01)
	Mean (SE)	27.61 (0.42)	2.31 (0.01)	9.58 (0.25)	7.91 (0.42)	2.44 (0.01)	1.64 (0.01)	0.08 (0.01)	0.05 (0.01)
Time × treatment	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)
I × B	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)
I × H	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)
I × S	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)
P × B	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)
P × H	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)
P × S	<i>p</i> value	0.155	0.315	0.414	0.125	0.169	0.365	0.284	0.058
	Mean (SE)	23.37 (0.67)	2.09 (0.07)	10.5 (0.3)	8.83 (0.35)	2.3 (0.09)	1.74 (0.05)	0.08 (0.01)	0.09 (0.01)
	Mean (SE)	22.91 (1.13)	2.04 (0.12)	9.89 (0.52)	9.78 (0.8)	1.67 (0.09)	0.19 (0.01)	0.08 (0.01)	0.01 (0)

Note: Mean values and standard errors (parentheses) in bold (*p* values ≤ 0.05) are for interpretation, according to ANOVA results. Means of significant treatment effects were separated using Tukey's tests. Time: I, 2-year-old aspen; P, 11-year-old aspen. Treatment: C, undisturbed control; B, burned; H, clearcut harvested; S, salvage logged.

Fig. 5. Treatment (undisturbed control, burned, clearcut harvested, and salvage logged) effects on (A) foliar N, (B) tree diameter at breast height (DBH), and (C) height at 10–11 years after disturbance (10/11 yr) for B and C and for both time periods considered (i.e., initial after disturbance and 10–11 years after disturbance) for A in western boreal mixedwood forest stands of Alberta. Different lowercase letters indicate significant differences ($P \leq 0.05$; Tukey's tests) among treatments combinations.

Jerabkova et al. 2011), in this study, after 10 or 11 years, $\text{NO}_3\text{-N}$ had increased in all treatments relative to the initial control, with highest values observed in the harvesting treatment. Immediate increases in extractable $\text{NH}_4\text{-N}$ under both fire (Wan et al. 2001) and harvesting (Staaf and Olsson 1994) have been observed on other boreal sites; however, in this study, after 10 or 11 years, $\text{NH}_4\text{-N}$ levels remained higher under harvesting than fire or salvage logging.

Harvesting did not result in significant N losses elsewhere (e.g., Simard et al. 2001), and in that study, harvesting increased potentially mineralizable N relative to fire and to undisturbed controls. Therefore, harvesting may stimulate a long-term increase in available N that is not observed under fire or undisturbed conditions. This may be due to greater N pools remaining on the site after harvest relative to after fire but warmer soil conditions than in the

control due to canopy opening. This increase in available N was reflected here in greater concentrations of foliar N in the harvesting treatment 10 or 11 years after harvest.

Although greater levels of foliar N in regeneration would be expected to correspond to increased regeneration growth, to date, this has not been realized. These stands are still in early stages of development; the influence of the soil nutritional environment on productivity may not be evident until canopy closure as other factors (e.g., microclimate) likely drive growth in the first decade of stand development (e.g., Thiffault et al. 2011; Ponder et al. 2012). The role of other site influences (e.g., soil physical properties, water balance, drainage, soil microbial processes) on regenerating stands requires further investigation.

Exchangeable base cations showed initial increase (K), decrease (Mg), or no difference (Ca) compared with the control, but after a

decade, exchangeable cations had converged and showed no treatment differences (Fig. 3B), and only in the case of K was the decadal level less than the initial control. Other work in the western boreal mixedwood forest shows that K and $\text{NH}_4\text{-N}$ exhibited the most definitive responses to harvesting (Kishchuk et al. 2014). These soils, formed on fine-textured glacial deposits under minimal acidifying deposition, do not warrant immediate concerns for base cation depletion. This is in contrast to base cation depletion under harvesting and acidifying conditions and following fire and salvage logging in eastern Canadian forests (Brais et al. 2000; Watmough and Dillon 2003; Duchesne and Houle 2006; Thiffault et al. 2007; Hazlett et al. 2011).

Cation exchange capacity has been shown to be an important large-scale and long-term indicator of western boreal mixedwood soil response to disturbance and stand development (Kishchuk et al. 2014). Forest floor CEC decreased under salvage logging over the 10-year period considered here (Fig. 3C). This could be due to changes in organic matter quantity but also quality, for example, in the nature and availability of functional groups, which are known to influence charge density per unit surface area of organic matter (Stevenson 1994). The incorporation into the soil of the highly condensed and aromatic C forms of charred organic material after fire may be critical to soil long-term exchangeable cation availability and retention capacity (Thiffault et al. 2008), a phenomenon that may be hampered on salvage-logged areas where sources of charred woody material are removed. Contrary to other studies (e.g., Simard et al. 2001; Neff et al. 2005), fire did not appear to increase long-term soil P availability. Fire has been shown to convert organic P to orthophosphate, and an increase in pH should also have a positive effect on P availability, but this pH-induced effect is usually short-lived (Certini 2005). By the time of the decadal remeasurement, extractable P levels had converged in the control, burned, and clearcut-harvest treatments to a level lower than the initial control condition; this pattern of decreasing P concentration over stand rotation has also been observed by Shrestha and Chen (2010). Increases in boreal soil pH following fire as observed here are well documented (Pietikäinen and Fritze 1993; Simard et al. 2001; Gómez-Rey and González-Prieto 2013); however, it is the influence of pH on properties such as soil nutrient availability that define its utility as a response parameter.

Increased forest floor extractable S over time under undisturbed conditions and low atmospheric S deposition has been observed elsewhere in the western boreal mixedwood forest (Kishchuk et al. 2014) and may be related to increased solubilization under stand development. Initial increases in extractable S in the burn and salvage-logged treatments may also be related to solubilization of both inorganic and organic S forms. However, the long-term decrease in extractable S under salvage logging could be related to persistent organic matter loss (Kishchuk and Brockley 2002).

Differences in mineral soil properties under disturbance were less evident than for forest floor. Where present, they occurred immediately after disturbance (e.g., available $\text{NH}_4\text{-N}$, extractable S, exchangeable K, and C and N content) but did not persist over time, results that are similar to those observed by Chen and Shrestha (2012). Johnson and Curtis (2001) noted that the influence of harvesting on the mineral soil was generally minimal. Moreover, the heat of wildfire often does not allow the temperature of the mineral layers to reach levels high enough to trigger significant changes in physical and chemical properties (DeBano 2000; Neff et al. 2005). At the site level, the mineral soil therefore appears to be buffered from acute changes occurring in the forest floor caused by disturbances (e.g., Thiffault et al. 2011).

Ecological and management implications

The properties most clearly affected under salvage logging are related to soil organic matter and C. The large removal of forest floor during fire, coupled with the subsequent removal of stand-

ing woody biomass has substantially reduced the C pool at this site. We have demonstrated the incremental effects of postfire salvage logging on boreal mixedwood soils beyond that of either fire or harvesting alone. This interactive effect of multiple disturbances occurring within a short time period demonstrates that the compound effects are greater than those of the individual disturbances (D'Amato et al. 2011; Bradford et al. 2012).

At 10 years after harvest, forest floor properties under clearcut harvesting, including depth, C content, pH, extractable $\text{NH}_4\text{-N}$, and extractable S, were not different from the control condition. In fact, harvesting alone resulted in long-term increases in available soil N, although, to date, this has not been reflected in regeneration productivity. This indicates that clearcut harvesting in these productive mixedwood stands has not resulted in adverse long-term soil changes. In contrast, changes in soil organic matter properties under salvage logging have persisted over 10 years. Relationships among these properties and stand productivity before and after canopy closure require further investigation.

Anthropogenic and naturally occurring impacts in the Canadian boreal forest are becoming more difficult to separate in, for example, the influence of changing climate on fire occurrence (Price et al. 2013). These interactions increase challenges in predicting impacts and require strategies for managing forested lands that take into account the direction, magnitude, and duration of compound effects. Results such as those that we present here provide empirical data that can contribute to longer term projections for western boreal forests. Management of these forests going forward will require a degree of flexibility that can accommodate mixedwood ecology in the context of the land-use pressures, disturbance, and changing climate increasingly coming to bear in Canada's boreal forests. With forest management for sustainable fibre supply remaining a core activity of this region (e.g., Keddy et al. 2013), opportunities exist to anticipate change and implement practices at the intersection of ecological and management trajectories that are aimed at maintaining or restoring a healthy and productive managed forest.

Conclusions

We have quantified persistent (decadal) soil changes resulting from specific disturbance types in the western boreal mixedwood forest. We have demonstrated that although many short-term changes in soil properties occur as a result of these disturbances, not all persist over the long term. Postfire salvage logging, as a compound disturbance, was shown to have an incremental effect on soil properties beyond the single disturbances with which it was compared, with the effects of postfire salvage logging persisting over a 10-year period. Clearcut harvesting did not result in adverse changes in soil properties over the decadal period. Instead, harvesting increased available soil and foliar nutrients. The profound reduction in soil C pools under severe fire and salvage logging, with a lower rate of recovery under salvage logging, has demonstrated differences in the effects of single and multiple disturbances on soil organic matter at these sites.

Our results show differences between short- and long-term responses to disturbance, among single natural and anthropogenic disturbances, and between single and compound disturbances in the western boreal mixedwood forest. The distinction between natural and anthropogenic disturbances is becoming increasingly difficult to delineate. We are facing an array of changes in the type, intensity, and frequency of disturbances at different scales. These changes require that we continue to develop and apply empirical knowledge of disturbance impacts to aid in projecting recovery of our forests and the development of future management practices.

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